

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

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Total Cells: 2000 | Model Mode: CLINICAL | Avg BioAge: N/A

1. Introduction

Cellular reprogramming using transcription factors has revolutionized regenerative medicine, allowing for the derivation of pluripotent stem cells from somatic cells. This experiment aimed to evaluate the efficacy of OSKM (Oct4, Sox2, Klf4, and c-Myc) transfection in inducing pluripotency in a somatic cell population.

2. Materials and Methods

The experiment involved the transfection of somatic cells with OSKM factors, performed twice to enhance reprogramming efficiency. Computational analysis included two rounds of single-cell variational inference (scVI) to assess cellular identity and population dynamics.

Experimental Interventions:

- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

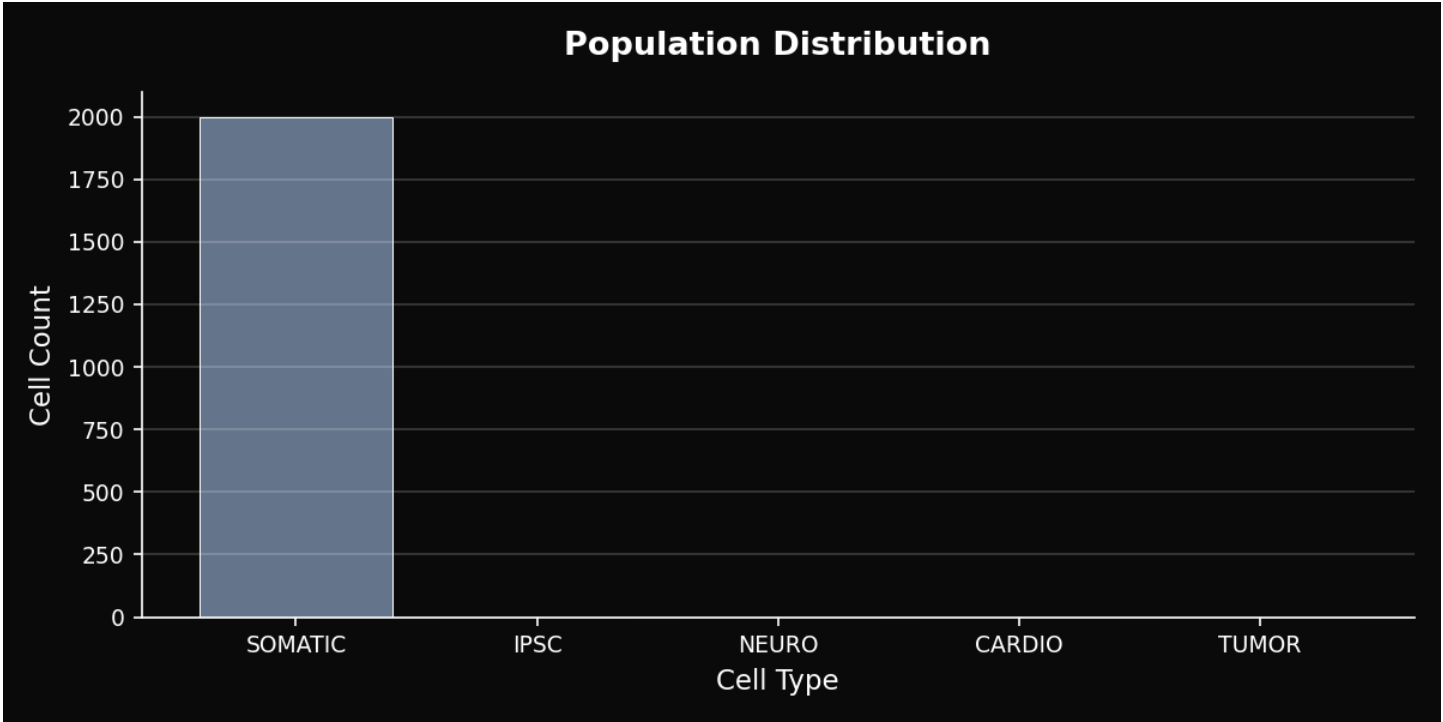
3. Results

Post-intervention, the final populations consisted of 2000 somatic cells, with no induced pluripotent stem cells (iPSCs), neural, cardiac, or tumor cells observed. The repeated OSKM transfection did not yield any detectable reprogramming to iPSCs or other differentiated cell types. scVI analyses confirmed the homogeneity of the somatic cell population with no emergence of alternative cell lineages.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	2000	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 2

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The absence of iPSC induction despite repeated OSKM transfections suggests potential inefficiencies in the reprogramming protocol or inherent resistance of the somatic cell type. These findings underscore the necessity to optimize transfection conditions or explore alternative reprogramming strategies. Understanding barriers to reprogramming in this context can inform improvements in protocols for regenerative applications.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.